

TEK4090

Introduction to Modern Control

Lecture 6: Feedforward Control

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Recap from Last Time

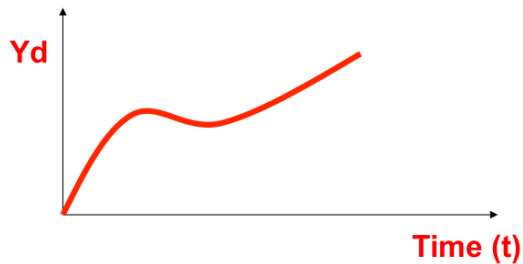
- ▶ Observability
- ▶ Estimators & State Estimation
- ▶ Optimal State Estimation: Kalman Filtering
- ▶ Applications to Spacecraft GNC

Reading Assignment

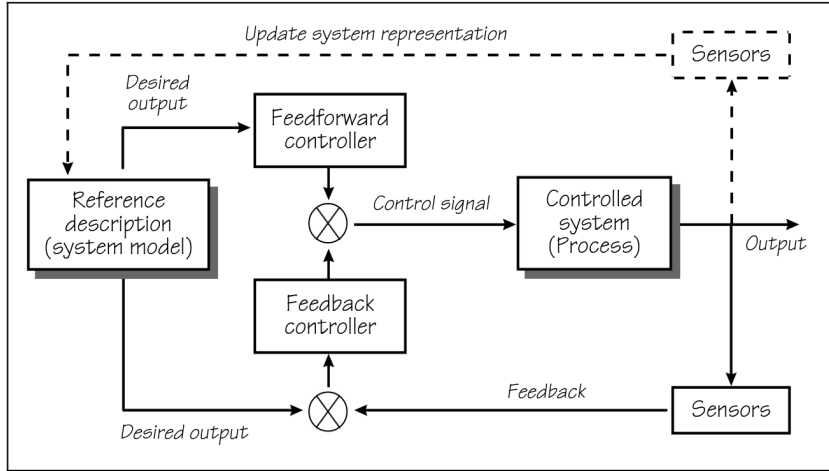
1. Notes by Santosh Devasia (University of Washington)

Motivation

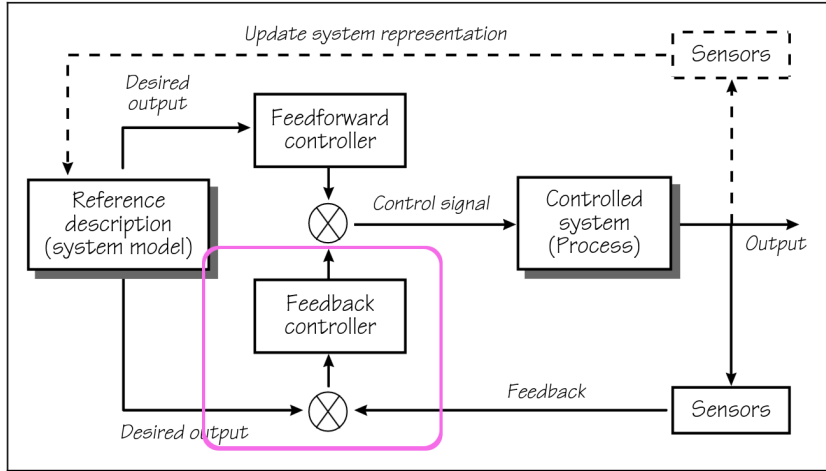
Goal: Find the input $u(t)$ that achieves a desired output trajectory $y(t)$



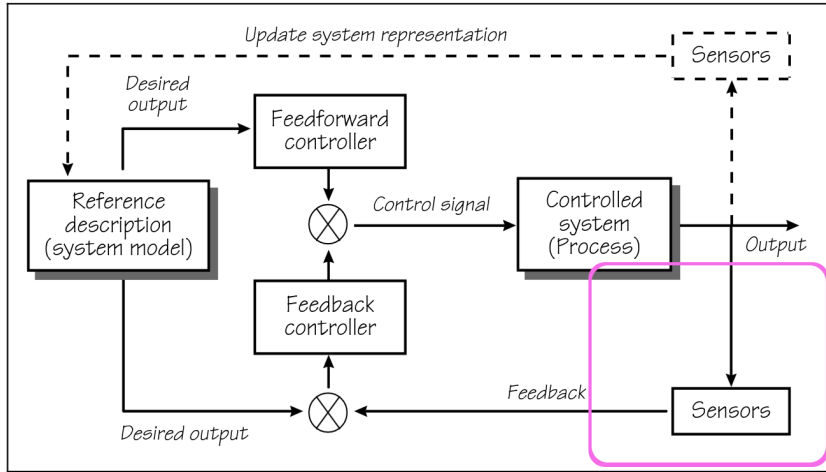
Control Block Diagram



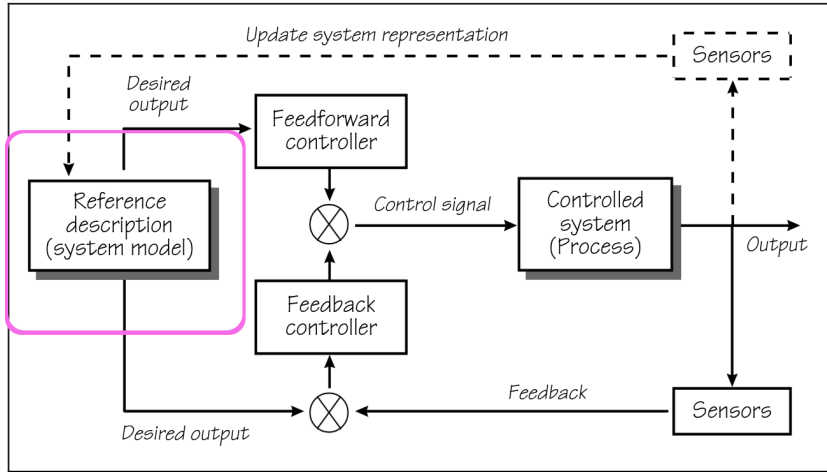
Control Block Diagram – Feedback



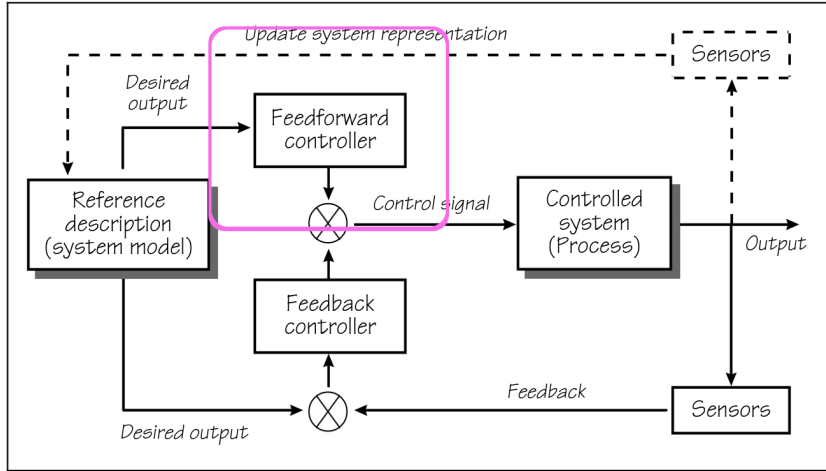
Control Block Diagram – Estimation



Control Block Diagram – System Identification



Control Block Diagram – Feedforward Control



System Inversion

Basic Example: Minimum-Phase SISO system

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s-1)}{(s-2)}$$
$$\implies Y(s) = \frac{(s-2)}{(s-1)} U(s)$$

Inverse Transfer Function

$$\frac{U(s)}{Y(s)} := G^{-1}(s) = \frac{(s-1)}{(s-2)}$$
$$\implies U(s) = \frac{(s-2)}{(s-1)} Y(s)$$

Note that both $G(s)$ and G^{-1} are **open-loop stable**

Setup

Linear SISO System

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$$u, y \in \mathbb{R}$$

Transfer Function

$$G(s) = C(sI - A)^{-1}B$$

Setup

Assumption: Well-defined relative degree.

The system has a well-defined relative degree r if,

$$\begin{aligned} CA^j B &= 0 & j < r - 1 \\ CA^{r-1} B &\neq 0 \end{aligned}$$

Setup

The relative degree r corresponds to the difference between the order of the denominator and the numerator of the transfer function

The Plan

Step 1: Differentiate Until the Output Appears

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Step 2: Decouple the State into Known/Unknown Parts

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Step 3: Write the Inverse Input Dynamics

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Step 4: Write the Internal Dynamics

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Step 4: Write the Internal Dynamics

Step 5: Complete the Inverse System

$$\dot{\eta}_{ref} = A_{inv}\eta_{ref} + B_{inv}\mathcal{Y}_d$$

$$u_{inv} = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$

$$A_{inv} = T_\eta [A - (BB_y^{-1}A_y)] T_r^{-1}$$

$$B_{inv} = \begin{bmatrix} T_\eta [A - (BB_y^{-1}A_y)] T_l^{-1} & \vdots & T_\eta BB_y^{-1} \end{bmatrix}$$

$$C_{inv} = -B_y^{-1}A_y T_r^{-1}$$

$$D_{inv} = \begin{bmatrix} -B_y^{-1}A_y T_l^{-1} & \vdots & B_y^{-1} \end{bmatrix}$$

$$\mathcal{Y}_d(t) = \begin{bmatrix} \xi_d(t) \\ y_d^{(r)}(t) \end{bmatrix}$$

Step 5.5: Solve the Inverse System

Step 6: For non-minimum phase systems

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Matlab Example

Non-Minimum Phase System

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$\longrightarrow$

$$\dot{\eta}_{ref} = A_{inv}\eta_{ref} + B_{inv}\mathcal{Y}_d$$

$$u_{inv} = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$

$\longrightarrow$

$$\dot{\eta}_s = A_s\eta_s + B_s\mathcal{Y}_d$$

$$\dot{\eta}_d = A_u\eta_u + B_u\mathcal{Y}_d$$

Focus: Internal Dynamics

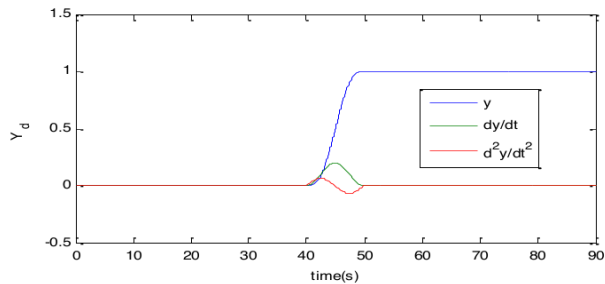
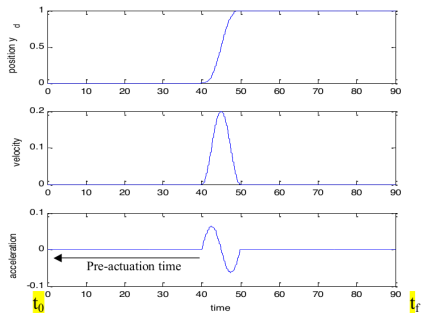
$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$A_s = -0.95$$

$$\dot{\eta}_d = A_u \eta_u + B_u \mathcal{Y}_d$$

$$A_u = 1.05$$

Desired Trajectory



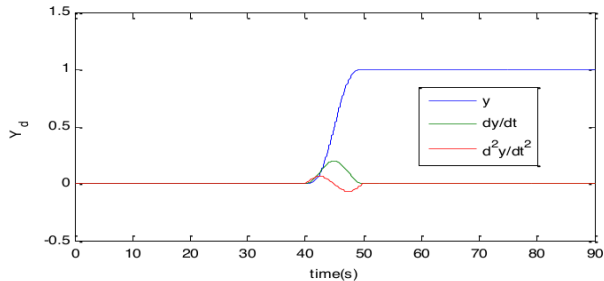
Solve Stable Subsystem

$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$Y_s = C_s \eta_s + D_s \mathcal{Y}_d$$

$$C_s = [1], \quad D_s = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Stable Subsystem

$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$Y_s = C_s \eta_s + D_s \mathcal{Y}_d$$

$$C_s = [1], \quad D_s = [0 \quad 0 \quad 0]$$

$$\mathcal{Y}_d = [y \quad \dot{y} \quad \ddot{y}]^T$$

Initial condition (from equilibrium:)

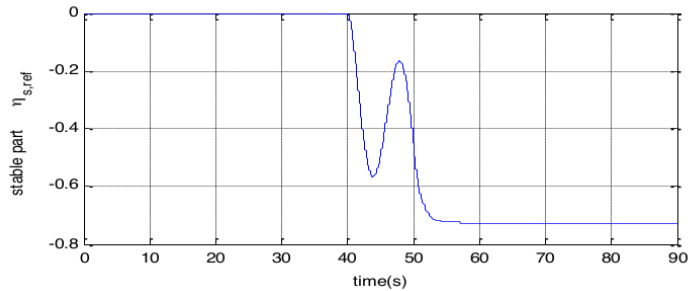
$$0 = A_s \eta_s + B_s \mathcal{Y}_d$$

$$\eta_s(t_0) = -A_s^{-1} B_s \mathcal{Y}_d(t_0)$$

$$\text{sys_s} = \text{ss}(A_s, B_s, C_s, D_s)$$

$$[\mathcal{Y}_d, \mathbf{t}, \eta_{s,ref}] = \text{lsim}(\text{sys_s}, \mathcal{Y}_d, \mathbf{t}, \eta_s(t_0))$$

Solve Stable Subsystem



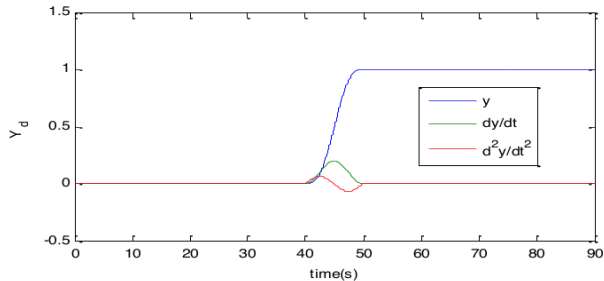
Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

$$C_u = [1], \quad D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

$$C_u = [1], \quad D_u = [0 \quad 0 \quad 0]$$

$$\mathcal{Y}_d = [y \quad \dot{y} \quad \ddot{y}]^T$$

Solve it backwards in time:

$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

Solve Unstable Subsystem

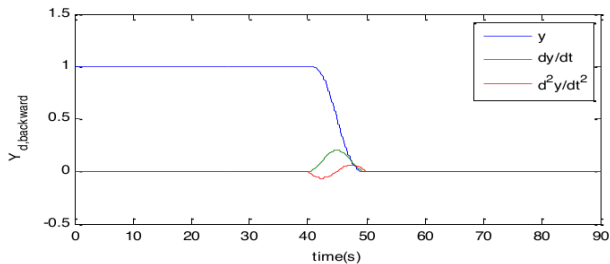
$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

$$C_u = [1], \quad D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$

Solve it backwards in time:



Solve Unstable Subsystem

Initial condition (from equilibrium:)

$$0 = A_u \eta_u + B_u \mathcal{Y}_d$$

$$\eta_u(t_f) = -A_u^{-1} B_u \mathcal{Y}_d(t_f)$$

$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

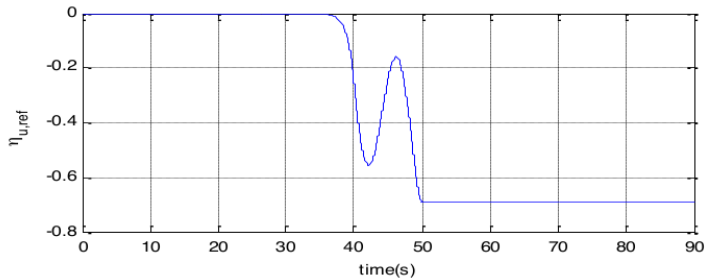
$$\text{sys_u_bw} = \text{ss}(-A_u, -B_u, C_u, D_u)$$

$$\mathcal{Y}_{d,bw} = \text{fliplr}(\mathcal{Y}_d)$$

$$[\mathcal{Y}_{ub}, t, \eta_{ub}] = \text{lsim}(\text{sys_u_bw}, \mathcal{Y}_{d,bw}, \eta_u(t_f))$$

$$\eta_{u,ref} = \text{fliplr}(\eta_u^T)$$

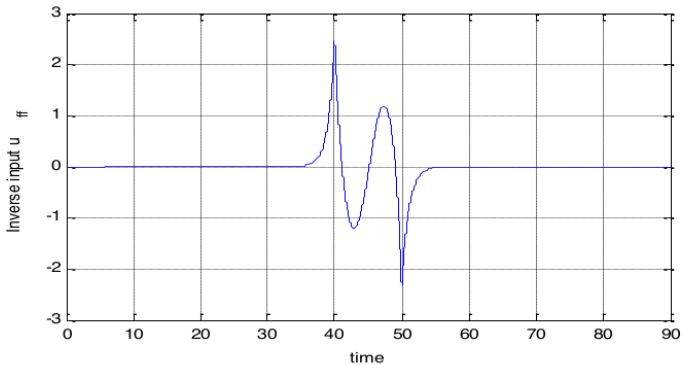
Solve Stable Subsystem



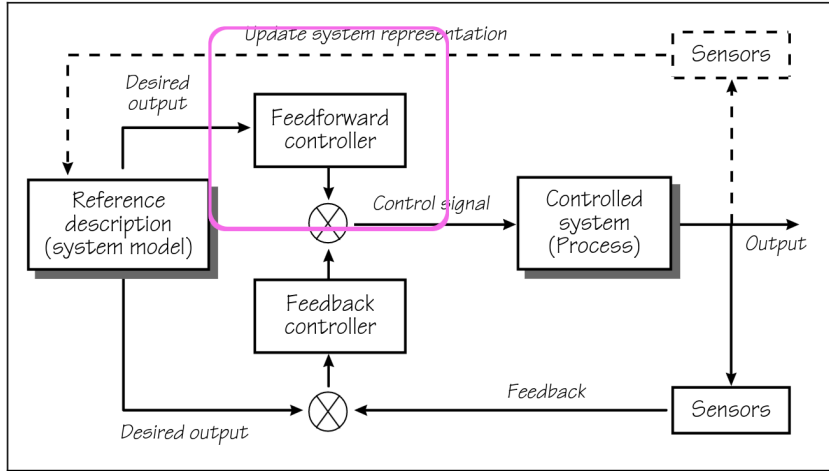
Find Reference Input

$$\eta_{ref} = T_{split} \begin{bmatrix} \eta_{s,ref} \\ \eta_{u,ref} \end{bmatrix}$$

$$u_{inv}(t) = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$



Today



Next Time

